CHEAT SHEET · NETWORKING FOR HACKERS · EP 01 · BEGINNER

How the Internet Works (and How Hackers Attack It)

One click, five stops. What each stop does, how attackers abuse it, how defenders stop them.

BitDark Labs

THINK LIKE A HACKER.
DEFEND LIKE A PRO.

01 The series cheat code

PACKET	= a postcard (from, to, message)
IP ADDRESS	= a street address
PORT	= an apartment number
DNS	= the phone book

02 Try it yourself

Find your IP address

Windows `ipconfig` Mac `ipconfig getifaddr en0`
Linux `ip a`

Starts with 192.168, 10. or 172.16–31? That's a private address.

Ask DNS yourself

`nslookup bitdark.net` The answer is the site's IP address.

03 Recap: 5 stops, attack vs defend

STOP	WHAT HAPPENS	HACKER	DEFENDER
1 **Your address**	Your device gets a private IP; the router shares one public IP (NAT).	Joins the same Wi-Fi and maps devices.	New router password, updates, care on public Wi-Fi.
2 **DNS**	The name bitdark.net is turned into an IP address.	Sends a fake answer to a fake site (DNS spoofing).	Encrypted DNS hides the question; DNSSEC signs the answer.
3 **Handshake**	SYN → SYN-ACK → ACK opens a connection to port 443.	Knocks on many ports to find open ones (port scan).	Firewall closes unused ports; alerts on scans.
4 **HTTPS**	The browser asks for the page; TLS locks it in an envelope.	Reads or changes plain HTTP (man in the middle).	HTTPS everywhere, HSTS, never click through warnings.
5 **Response**	The server answers with a status code and the page.	Reads detailed errors to learn software versions.	Simple error pages, details in private logs, patching.

04 New terms

Packet · a small chunk of data with a "from" and "to" address.
Private IP · works only inside your network (a room number).
Public IP · your network's address on the internet.
NAT · the router swaps private for public addresses.
DNS · turns names like bitdark.net into IP addresses.
DNS spoofing · a fake DNS answer that sends you to a fake site.
Port scan · checking many ports to see which are open.
Port · a numbered door on a server; HTTPS uses 443.
TCP handshake · SYN, SYN-ACK, ACK: a hello before data.
HTTP · the language browsers use to ask for pages.
TLS · the encryption behind HTTPS (the locked envelope).
Certificate · a website's ID card, checked by your browser.
HSTS · tells browsers to always use HTTPS for a site.

Status code · 3-digit result of a request:
`200 OK` `301 moved` `404 not found` `500 error`

Quiz: You type the right address but land on a fake bitdark.net. Which stop was attacked?
A · Your address B · DNS C · The handshake D · The response

Answer: B, DNS. A fake DNS answer sent you to the wrong IP address.

For education only. Only test systems you own or have permission to test.